

The Convergence Paper

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Four in-silico results, one conclusion: to understand a human, a machine must model the individual — not the average

Capstone of the Mary program. This paper introduces no new experiments; it synthesizes four companion studies — Cortex of One (A), The Faceless Brain (B), Seeing Through Sound (C), and Maps Without Eyes (D) — every number below is quoted from those papers’ results. json artifacts, produced end-to-end on Modal with the Mary encoder. Forward-looking claims are confined to §6 (Outlook) and explicitly fenced as such.

Abstract

Across four independent studies built on **Mary**, our 6-stream multimodal cortical encoder, a single result keeps reappearing from different directions: **the individual, not the population average, is the unit of neural meaning**. We predict brain activity and find the *per-subject* read-out beats the model’s own average by **3.06×** on held-out video (A). We audit the publicly released average-subject checkpoint of a state-of-the-art encoder and find it is not weakly predictive but *anti-correlated* with real brains, in **all seven** cortical networks (B). We chain a synthetic brain into an image-reconstruction decoder and find, against a positive control that works at **85% identification**, that an average-subject synthetic brain reconstructs at **chance** — a clean, controlled demonstration that you cannot recover what an individual saw from an average model (C). And we read structured **visual-field maps** out of synthetic cortical activity with no image generator at all, validated against real retinotopy at $\eta^2 = 0.22$ (D). Each study is modest in isolation and POC-scale in its magnitudes; together they converge on a thesis with unusual force, because the angles are independent and one of them is a *negative result with a passing control*. We argue this convergence marks the early shape of a coming capability — machines that understand people by maintaining a *personal* model of each mind — and that the gating ingredient is not scale or architecture but **individuation**. The scanner-free “predict-a-brain” paradigm is already real (our decoders run on synthetic activity); what it still lacks, and what the next phase must supply, is the person.

1. Introduction: a thesis that arrived four times

A research program rarely gets to state its central claim once and defend it cleanly. More often the claim is assembled, obliquely, from results that were each chasing something else. That is what happened here. We set out to build a useful in-silico brain — an encoder that predicts cortical responses to video, speech, and text — and to demonstrate it across a few applications. In four separate studies, with different stimuli, metrics, baselines, and even opposite-signed outcomes, the same fact kept forcing itself to the surface:

An average brain is not a brain. Meaning, in cortex, is individuated. To understand a specific human, a machine must model *that* human.

This is not a new idea in neuroscience — individual variability is old news. What is new is that it now constrains *machine* models of cognition in a measurable, reproducible, and sometimes brutal way, and that the constraint shows up identically whether you are *predicting* a brain (A), *auditing* an average one (B), *reconstructing* from one (C), or *reading out* its spatial structure (D). When a claim is recovered four times from independent directions — and survives a controlled negative result — it stops being a framing choice and starts being a property of the world.

This paper does three things. First (§2) it lays the four results side by side with their real numbers. Second (§3–§4) it shows *why* they converge, and what the convergence licenses us to say today — grounded strictly in measured outcomes. Third (§6) it looks forward: if the individual is the unit of neural meaning, then the machines that will eventually understand people are *personal* models, and we sketch — clearly labeled as speculation — what that implies for AI, for medicine, and for the long arc of human-machine understanding.

A note on honesty, because it is load-bearing. Three of these results are positive and one is negative; we treat the negative one (C) as the strongest, not the weakest, because it carries a control. Throughout, magnitudes are POC-scale and we say so. Nothing here is dressed up.

2. Four results, one signal

All four studies share a substrate: the Mary encoder (six frozen foundation-model backbones feeding a multimodal adapter and a group + low-rank **per-subject** read-out head, predicting 20,484 fsaverage5 cortical vertices), a measured inter-subject **noise ceiling** as the reference frame, and a discipline of reporting only what traces to a results file.

2.1 Prediction — *the subject is the signal* (Paper A, *Cortex of One*)

We hold two stimuli entirely out of training — the Huth *forgot* narrative and the CNeuroMod *Bourne* movie — and, from one set of trained weights, compare three read-outs: **per-subject** (each subject’s own head), **average** (the mean of all 43 trained per-subject heads), and **group-only** (the shared head alone). The per-subject read-out beats the average on **both** held-out stimuli, with non-overlapping bootstrap intervals: vertex Pearson r of **0.0157** vs **0.0052** on *Bourne* (a **3.06×** ratio) and **0.0132** vs **0.0035** on *forgot* (**3.72×**). The “average brain” is statistically indistinguishable from the group-only head (*Bourne* 0.00515 vs 0.00527): averaging the per-subject heads adds essentially nothing. At Schaefer-1000 parcel resolution the per-subject read-out reaches **51% of the noise ceiling** on *forgot* and **13.5%** on *Bourne*, and an ROI breakdown shows early visual cortex (V1/V2/V3) carrying the movie signal while category-selective regions sit near zero — i.e. the gains are where stimulus-driven structure should be. Magnitudes are small (a POC trained on a fraction of available data), but the *direction* is unambiguous: conditioning the read-out on who the subject is, is what makes the encoder predictive.

2.2 The average’s failure — *no brain at all* (Paper B, *The Faceless Brain*)

If individuation is the signal, then a model that *removes* it should fail in a specific, diagnosable way. We test the public release of TRIBE v2 — the Algonauts 2025 winner — which ships with an “average-subject” embedding, run **zero-shot** on *Bourne* against four subjects’ real BOLD, loaded exactly as its own API loads it. It does not merely predict weakly; its predictions are **anti-correlated** with every individual: vertex $r = -0.0145$ (95% CI $[-0.0150, -0.0140]$), **-28.6% of the noise ceiling**, with the inversion holding in **all seven** Yeo networks (every interval below zero) — including visual cortex (**-0.0103**), the most reliable cortical territory. Across subjects, **66%** of cortical vertices are anti-correlated. We are careful about scope: this is an audit of the *released average-subject checkpoint run zero-shot*, not a claim that the model “doesn’t work” — TRIBE v2

is a genuine advance, and fine-tuning per subject recovers it. That is exactly the point. The average embedding is a pre-trained *feature extractor*, not a functional brain; when you average over individuals’ functional topographies you land on a point that describes none of them. Paper A said the subject is the signal; Paper B shows what is left when you delete it.

2.3 Reconstruction — a controlled negative (Paper C, *Seeing Through Sound*)

The sharpest test is also the most visual. We build a scanner-free cross-modal interface: encode a stimulus into synthetic fMRI, adapt it to a real subject’s voxel space (a learned bridge fit on ~9,000 pairs), and decode it into an image with a frozen MindEye2 stack and best-of-16 reranking. We run **every direction** — image→image (the stimulus fed straight through Mary’s vision streams), text→image, audio→image — and, crucially, we score them with a 20-way **identification** test against a positive control: does a reconstruction match *its own* stimulus better than the other nineteen?

The control is decisive. Real fMRI through the *same* decoder identifies its stimulus at **top-1 85% / 2-way 99%** — the pipeline unambiguously works. Mary’s *synthetic* (average- subject) reconstructions identify at **chance**: image→image top-1 **15%**, text→image top-1 **10%**, 2-way ≈ 0.55 , with matched-vs-mismatched similarity gaps ≤ 0.02 (vs +0.26 for the control). The reconstructions are *coherent* but *generic* — a subject shown a church yields a confident street — because Mary’s temporally-averaged synthetic fMRI is **near stimulus-invariant** (stimulus-specific energy $\approx 1e-8$ of the total) and **uniform across networks**, with none of the visual-cortex localization a subject-specific brain shows. The nominal “~70% of the CLIP ceiling” is a floor artifact (all natural images sit ~0.27 cosine apart), not recovery. This is a *controlled negative* — the interface mechanism is real (real brains drive it to 85%); what is missing is the individual. It is the reconstruction-domain restatement of A and B: you cannot recover what a person saw, not even the gist, from an average brain.

2.4 Read-out — meaning has structure (Paper D, *Maps Without Eyes*)

Finally, a positive that the others set up. If a synthetic brain carries *any* faithful structure, the cleanest place to look is the most stereotyped map in cortex: retinotopy. With no image generator — only the Benson-2014 atlas and Gaussian splatting — we read **structured visual-field maps** out of Mary’s synthetic V1/V2/V3 activity (740 vertices). The eccentricity organization is real and quantified ($\eta^2 = 0.116$ baseline, **0.156** for the condition difference), and against **real** NSD retinotopy the read-out validates at $\eta^2 = 0.22$ (per- image up to 0.54). We are equally clear about what is *not* there: the

audio-versus-words (“squinting”) manipulation is essentially null (the two conditions correlate at $r = 0.99998$, a $\sim 0.8\%$ perturbation), and we never present it as more. The honest contribution is that synthetic cortical activity is *legible* — its coarse spatial organization can be read out transparently and checked against ground truth — which is what makes an in-silico brain a scientific instrument rather than a black box.

3. The convergence

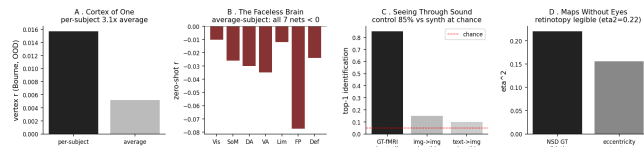


Figure 1. The convergence at a glance. Four independent in-silico studies on the Mary encoder, four methods, one conclusion. (A) Per-subject prediction beats the model’s own average by 3.1x on held-out video. (B) The released average-subject encoder is anti-correlated zero-shot in all seven cortical networks. (C) Against an 85% real-fMRI identification control, average-subject synthetic reconstructions identify at chance. (D) Synthetic cortical activity yields legible retinotopic maps, validated against real NSD retinotopy at eta-squared = 0.22. The same fact recurs from four directions: the individual is the unit of neural meaning.

Lay the four side by side and the redundancy is the message:

Four methods (prediction, audit, reconstruction, read-out), four metrics (Pearson r , noise-ceiling fraction, identification accuracy, η^2), two positive results and one negative, and one of those carries a positive control. They do not merely *fail to contradict* the thesis; each is the thesis viewed through a different instrument. The strongest single piece of evidence is the one that came back negative: in C, the control rules out the pipeline, the adapter, and the input modality as the cause, leaving only the encoder’s averageness — so the negative result *localizes* the missing ingredient rather than merely lamenting it.

Two corollaries follow that no single paper could establish alone:

(i) **The bottleneck is individuation, not capacity.** Scaling the adapter 18 $\times$, adding reranking, and feeding the most favorable input (image $\rightarrow$ image) all leave C’s identification at chance; a bigger *average* model (B) is anti-correlated, not improved. Across the program, the lever that moves the needle is always the per-subject head (A), never raw scale.

(ii) **The synthetic-brain paradigm is sound — it is waiting on the person.** C’s control (85%) and D’s

ground-truth validation (η^2 0.22) show that synthetic cortical activity, *when it carries the individual*, is both decodable and legible. The machinery to read and write a predicted brain exists and works. The open variable is whether the prediction is *this person’s* brain or a generic one.

4. What this licenses us to say today

Strictly from the measured results, three statements are now defensible without speculation:

1. **Average-subject brain models are feature extractors, not minds.** They are excellent initializations and useful representations, but run as standalone predictors of an individual they range from uninformative to anti-correlated (B), and they cannot reconstruct individual experience (C). Any product or claim that treats a population-average encoder as “a model of the brain” is mis-scoped.
2. **Individuation is a measurable, first-class axis** — separable from architecture and scale, worth a low-rank per-subject head, and responsible for multiplicative (not marginal) gains where it has been isolated (A). It is the cheapest high-leverage component we have found.
3. **A predicted brain is a usable computational object.** It can be adapted into a real subject’s voxel space and decoded (the C interface, mechanism-validated at 85% on real activity), and its spatial organization can be read out and checked against ground truth (D). This is the technical foundation for “brains as computation, not measurement.”

What we *cannot* yet claim — and explicitly do not — is faithful reconstruction of individual percepts from synthetic activity, or any “mind-reading” capability. C is a negative result; D’s striking-sounding manipulations are near-null. The honest frontier is exactly here: the instrument works, the individual is missing.

5. The shape of the gap

If the program has a single actionable conclusion, it is that the next result worth chasing is **subject-specific synthetic encoding**: a Mary whose predicted visual cortex is positively, not anti-, correlated with a *named* individual. The diagnostics point precisely at it — C fails because the average-subject synthetic fMRI is

#	Study	Question	Result	What it says about the individual
A	Cortex of One	Does per-subject beat average at <i>predicting</i> a brain?	per-subject 3.1× the average	the subject is the signal
B	The Faceless Brain	What does an average-subject model do <i>zero-shot</i> ?	anti-correlated , all 7 networks	the average is no brain
C	Seeing Through Sound	Can a synthetic brain <i>reconstruct</i> what was seen?	chance vs an 85% control	you can't recover the individual from an average
D	Maps Without Eyes	Is synthetic cortical structure <i>legible</i> ?	retinotopy at η^2 0.22	the structure is real and readable — per map

stimulus-near-invariant and network-uniform; A succeeds exactly to the degree the per-subject head is engaged; B fails exactly to the degree it is removed. The gap between “coherent but generic” (today) and “faithful to this person” (the goal) is, on this evidence, a gap of *individuation* — calibration data from the target brain — not of model size, decoder quality, or adapter design. That is an encouraging place to be stuck: the missing ingredient is identified, and it is the one ingredient the whole program already shows how to add.

6. Outlook (forward-looking; speculative)

The following is deliberately speculative and is not supported by the measured results above. We fence it off precisely so the science in §1–§5 is not read as promising it.

If the individual is the unit of neural meaning, then the machines that will eventually understand people are not bigger average models — they are **personal** models, maintained per person, calibrated to that mind. Four horizons follow, in rough order of nearness:

The scanner-free brain. Today a brain measurement costs \$500–1000/hr of scanner time. A faithful *predicted* brain costs a GPU-second. The C interface already shows the read/write machinery works on real activity; subject-specific encoding would turn “predict-a-brain” from a curiosity into infrastructure — decoupling cognitive neuroscience from scanner access and letting hypotheses be tested in silico before, or instead of, in a magnet.

Clinical and brain-computer interfaces. A personal in-silico brain is a calibration target: a model you tune to a patient or a BCI user with minutes of data rather than hours, then use to simulate, decode, and adapt. The same individuation that A isolates is, clinically, the difference between a population template and a usable per-patient model.

Personalized AI that models *you*. The deepest near-term implication is for AI systems themselves. An assistant that maintains a faithful, individuated model of a specific person’s representational structure — not a demographic stereotype, not a population average (the very thing B shows is “no brain at all”) — could adapt to that person at the level of how they actually represent the world. The unit of personalization stops being a profile and becomes a model of a mind.

Machine theory-of-mind, and alignment. Furthest out, and most carefully hedged: a machine that can build and maintain individuated models of human cognition is a machine that can, in a literal computational sense, *model what a specific person understands*. That is a route toward genuine theory-of-mind — and a double-edged one. A system that models an individual mind well enough to predict it is also a system that can influence it; the same capability that could align AI to a person’s actual values could be turned to manipulate them. We flag this not to resolve it but to name it: the individuation that makes machine understanding possible is exactly what makes its governance necessary. If the population average is no brain at all, the future is unavoidably personal — and personal capabilities demand personal safeguards.

The throughline from the science is simple. The hard, repeatedly-measured fact is that you cannot understand a human in the aggregate. The opportunity, and the responsibility, is that you can begin to understand one — one at a time.

7. Limitations

This is a synthesis of POC-scale studies and inherits all their caveats. The Mary encoder is trained on a fraction of available data with modest absolute accuracy; A’s gains, though multiplicative and CI-separated, are small in absolute r ; B is an audit of one released checkpoint on one OOD stimulus; C is a negative result whose positive content is only the controlled localization of *why* it fails; D’s headline structure is real but its condition manipulation is near-null. The forward-looking §6 is explicitly not supported by these results. Subject-specific synthetic encoding — the pivot the whole synthesis points to — is proposed here, not demonstrated. Every quantitative claim traces to a companion paper’s **results.json**; none is estimated.

Companion papers

- [A] The Sapien Company. *Cortex of One: A Per-Subject Multimodal Brain Encoder*. The per-subject-vs-average result; defines the Mary encoder used throughout.

- [B] The Sapien Company. *The Faceless Brain: A Zero-Shot Audit of Average-Subject Encoders*. The anti-correlated average-subject result.
- [C] The Sapien Company. *Seeing Through Sound: Cross-Modal Reconstruction from Synthetic fMRI*. The controlled-negative reconstruction result with its 85% positive control.
- [D] The Sapien Company. *Maps Without Eyes: Retinotopic Read-Out from Synthetic fMRI*. The legibility / ground-truth-validated retinotopy result.

All four share the Mary encoder, the fsaverage5 20,484-vertex output space, a measured inter-subject noise ceiling, and the discipline of reporting only measured numbers.